Cellular Response to Injury
Adaptation & Reversible cell Injury

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SGL 1

Dr. Payman Anwar Rashid
Assistant Professor of Pathology
Consultant Histopathologist
Department of Anatomy and Pathology
College of Medicine
Hawler Medical University

Lecture Outline:

* Overview of cellular response to injury.
* Cellular adaptation to injury.
    * Hypertrophy.
    * Hyperplasia.
    * Atrophy.
    * Metaplasia.
* Morphology of Reversible Injury

Objectives:

1. To understand concept of cellular response to stress.
2. To explain types of cellular adaptation.
3. To describe morphology of reversible cell injury.

Cellular Response To Injury

- Normal cell function requires a balance between physiologic demands, the structure of cell and metabolic capacity; the result is a steady state, or homeostasis.

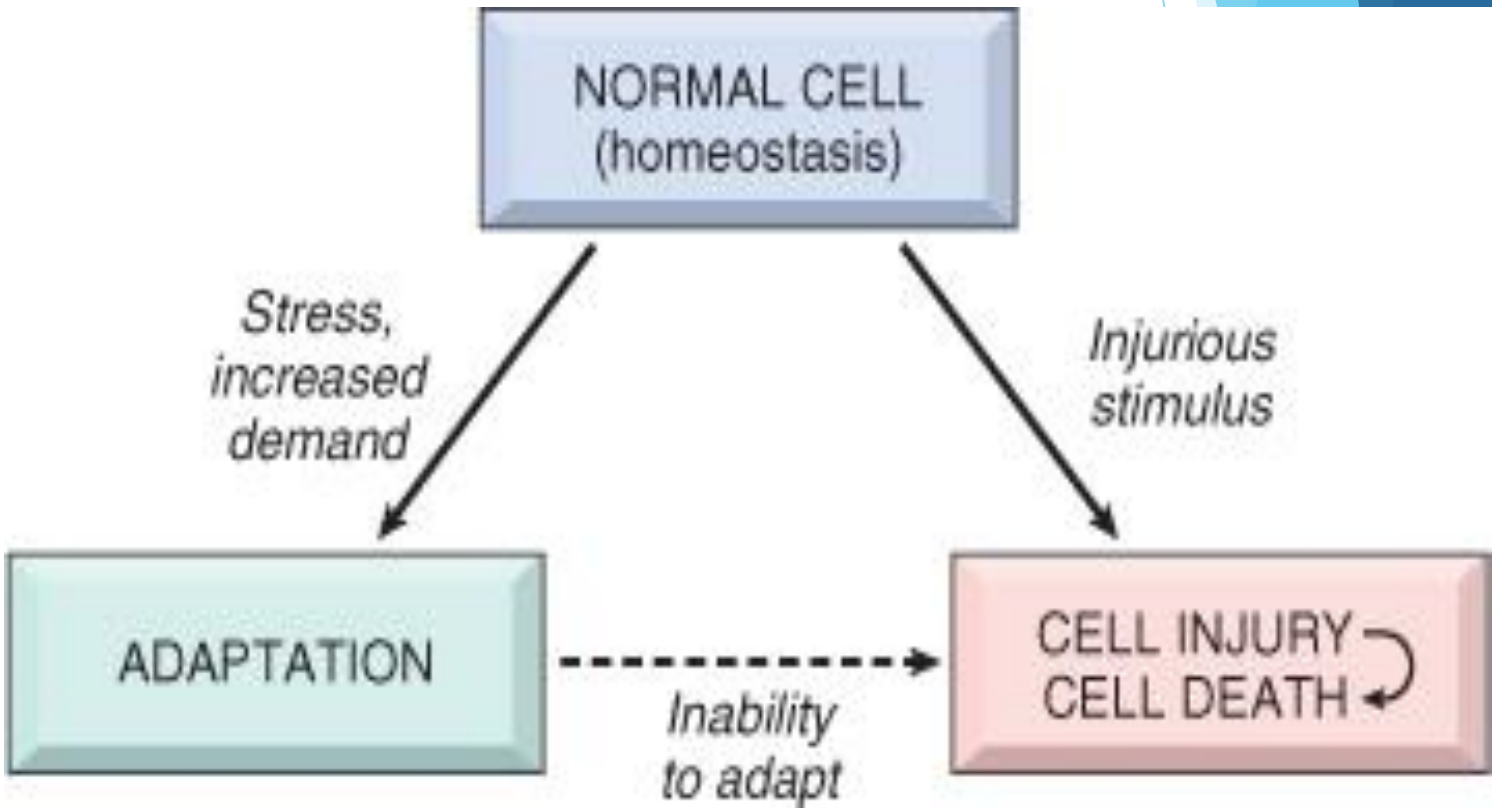
Cells can alter their functional state in response to modest stress to maintain the steady state.

More excessive physiologic stresses or adverse pathologic stimuli (injury), result in:

1. Adaptation.

2. Reversible injury.

3. Irreversible injury and cell death.



- **Adaptation:**

occurs when physiologic or pathologic stressors induce a new state that change the cell but otherwise preserves its viability in the face of the exogenous stimuli.

These changes include:

- 1. **Hyperplasia** (increased cell number).
- 2. **Hypertrophy** (increased cell mass).
- 3. **Atrophy** (decreased cell mass).
- 4. **Metaplasia**. (change from one mature cell type to another).

- **Hyperplasia:**

is an increase in the number of cells in an organ or tissue. It is usually accompanied by hypertrophy. Hyperplasia can occur only with cells capable of synthesizing DNA (such as epithelial, hematopoietic, and connective tissue cells).

Nerve, cardiac, and skeletal muscle cells have no capacity for hyperplastic growth, so muscle cells undergo pure hypertrophy when stimulated by increased functional load or hormones.

Hyperplasia can be physiologic or pathologic

- **1. Physiologic hyperplasia:**

* Hormonal hyperplasia, exemplified by the proliferation of the glandular epithelium of the female breast at puberty and during pregnancy.

* Compensatory hyperplasia (e.g. hyperplasia of the liver after partial hepatectomy).

2. Pathologic hyperplasia:

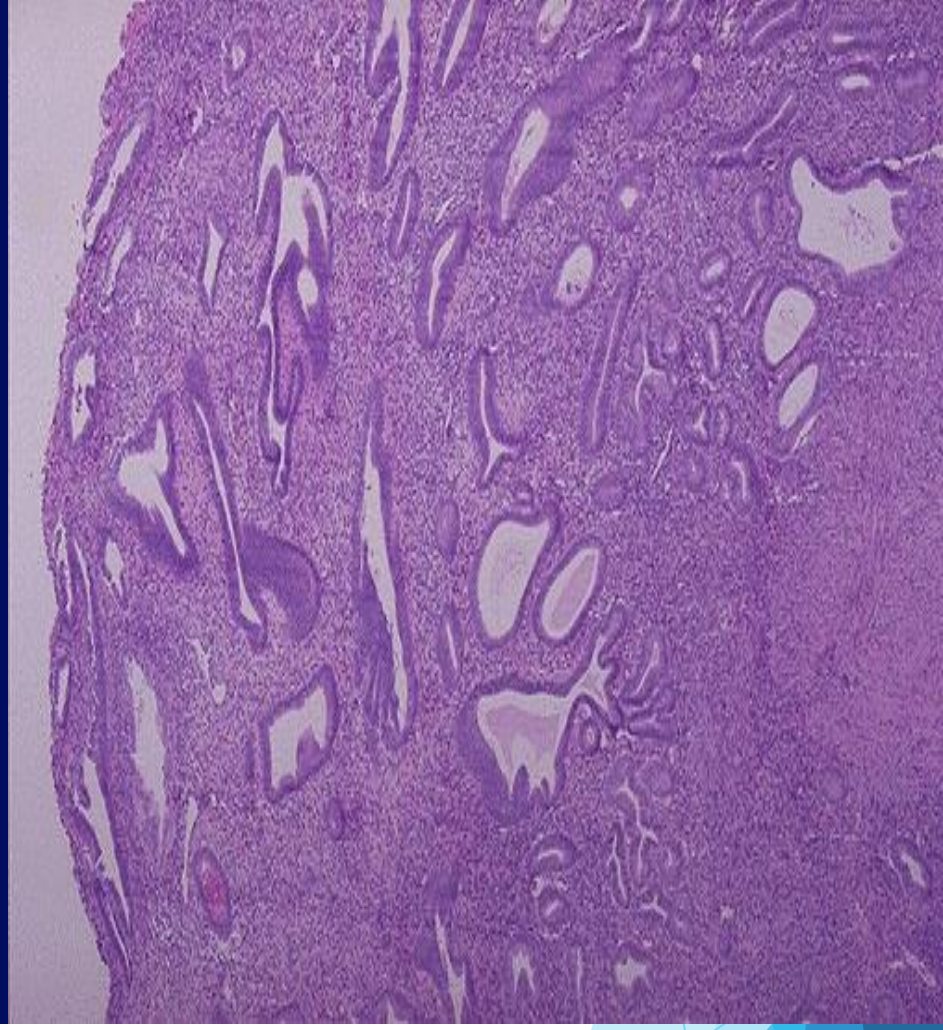
* Excessive hormonal stimulation (e.g. hyperestrinism and atypical endomaterial hyperplasia).

* Effects of locally produced growth factors on target cells (e.g. proliferation of connective tissue cells in wound healing).



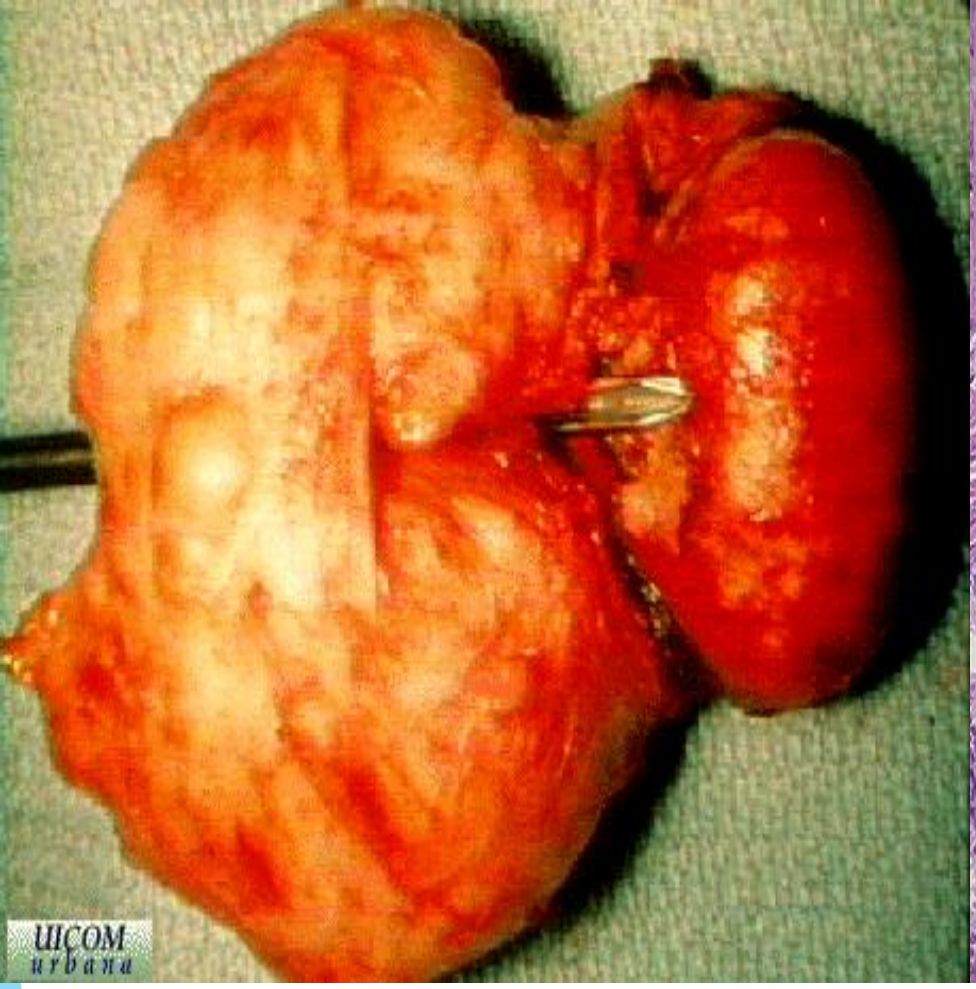
Endometrial hyperplasia

(Pathologic hormonal stimulation)

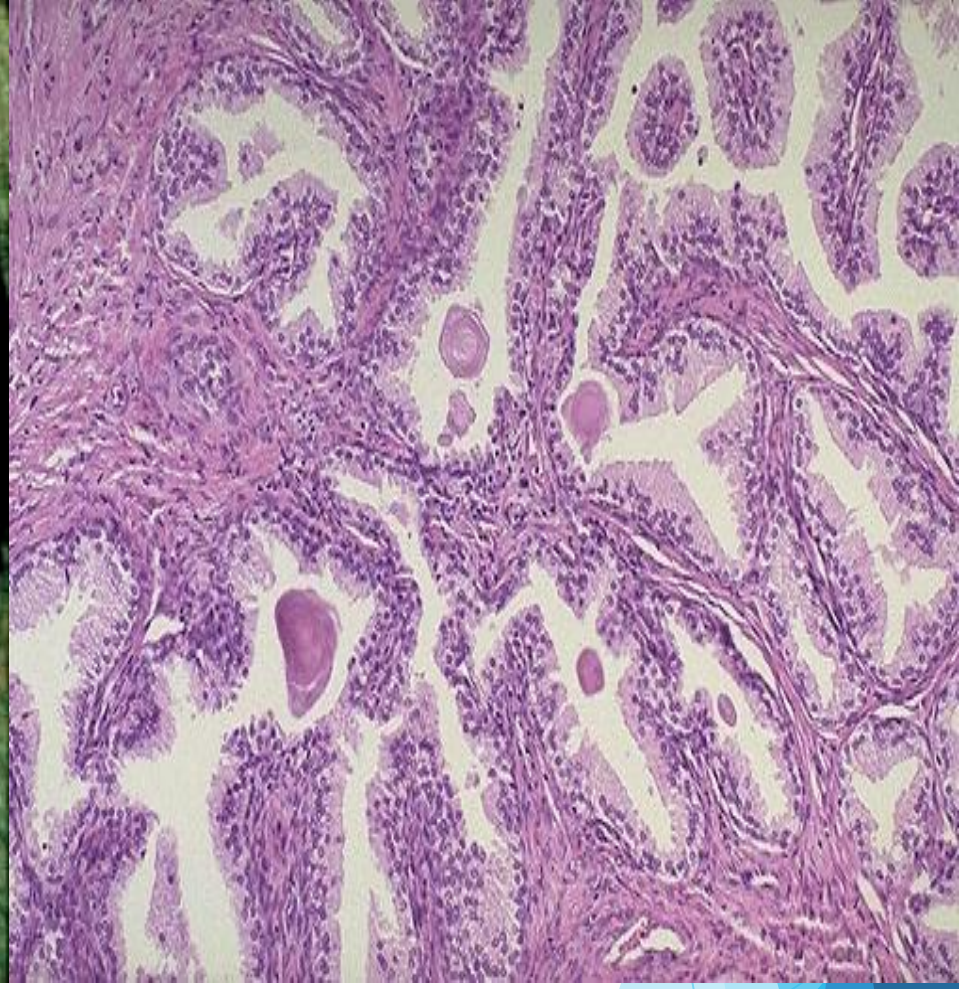


Endometrial hyperplasia

(microscopical appearance)



Benign prostatic hyperplasia
Pathologic hormonal hyperplasia



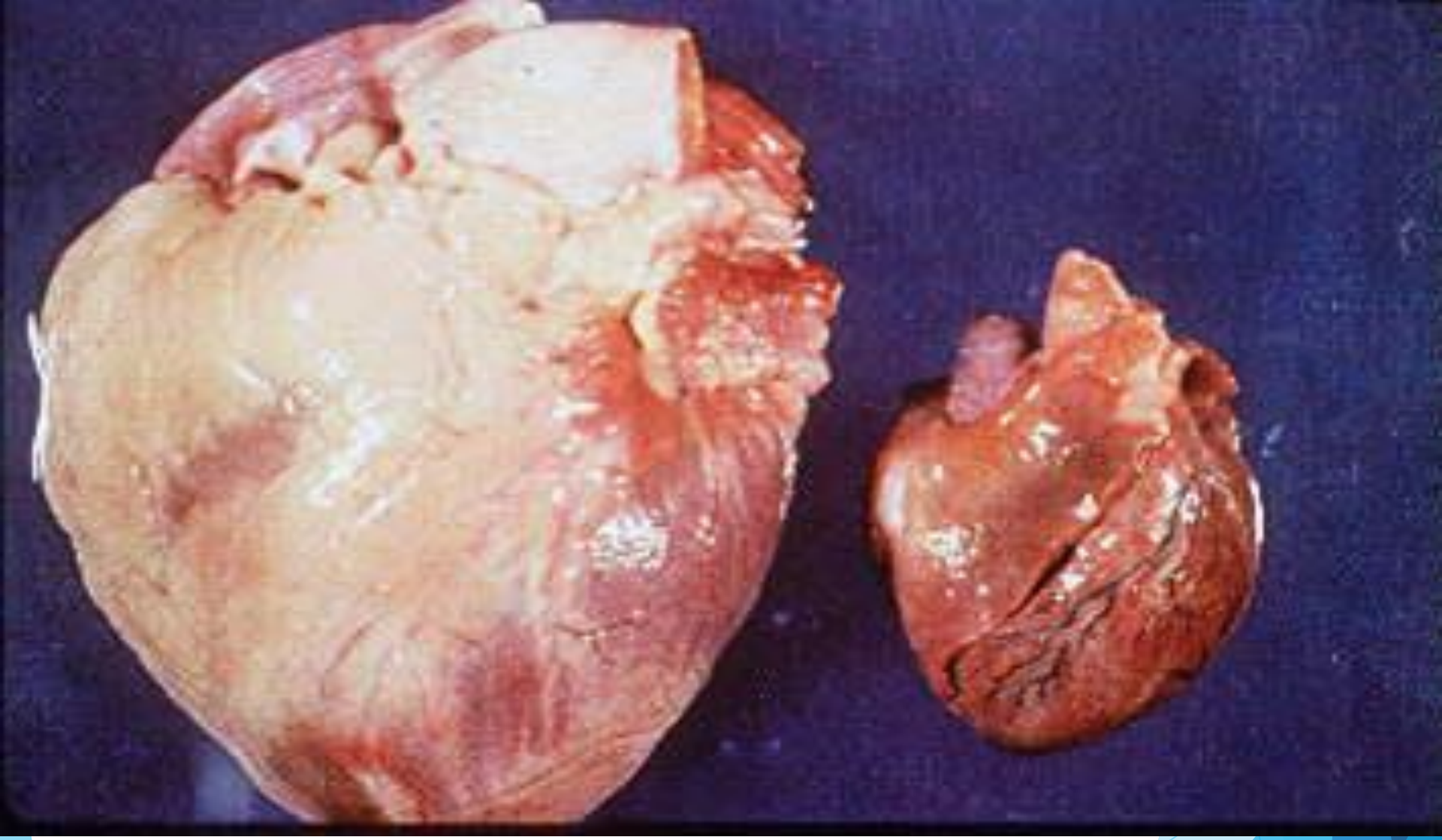
Microscopical appearance of BPH

- In pathologic hyperplasia if the stimulus removed the hyperplasia disappears.

* However pathologic hyperplasia constitute a fertile soil in which cancerous proliferation may eventually arise. E.g. atypical endometrial hyperplasia which are precursors of cancer of endometrium.

Hypertrophy:

- is an increase in the size of cells which lead to increase in size of organ.
- Hypertrophy can be physiologic or pathologic and is caused by :
- 1. Increased functional demand e.g. hypertrophy of striated muscles in muscle builders (physiologic) or of cardiac muscle in cardiac disease (pathologic).
- 2. Specific hormonal stimulation e.g. uterine hypertrophy during pregnancy.



Heart hypertrophy due to work overload (as in hypertension) compared with normal heart (Rt.side) for the same age & sex.

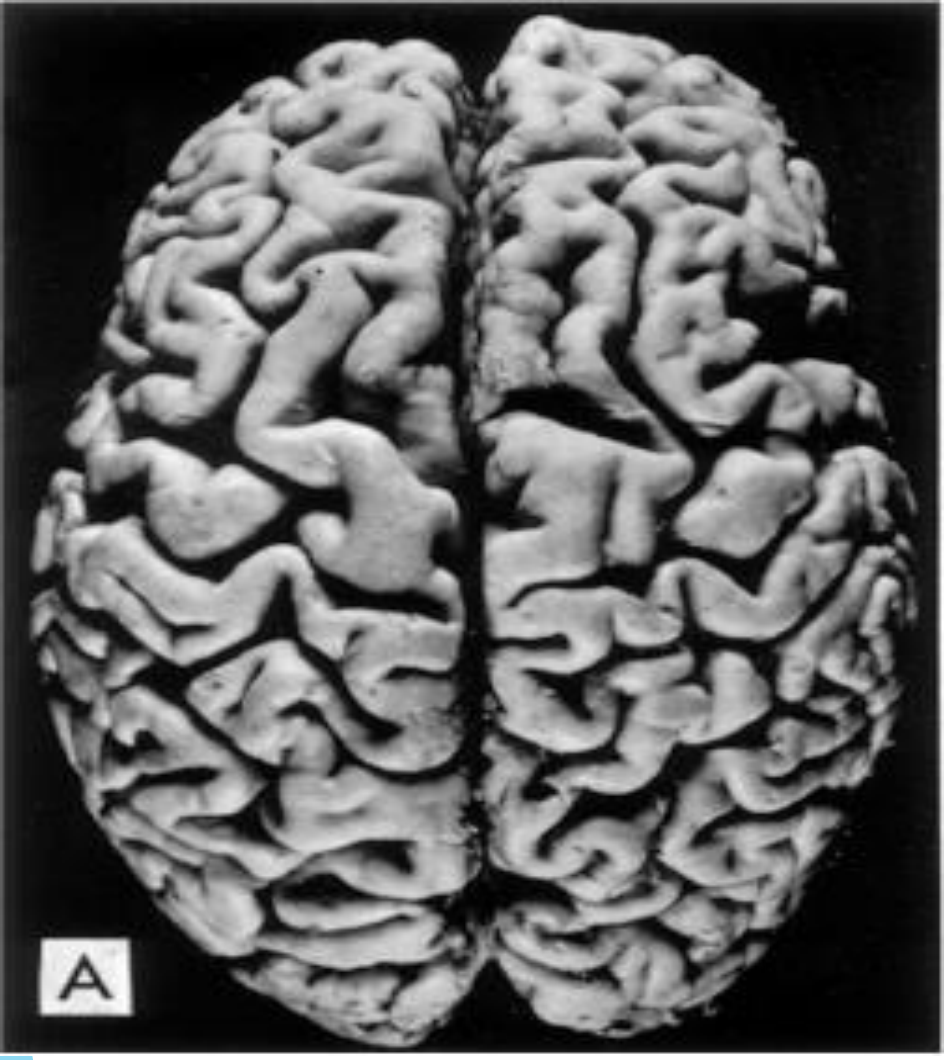


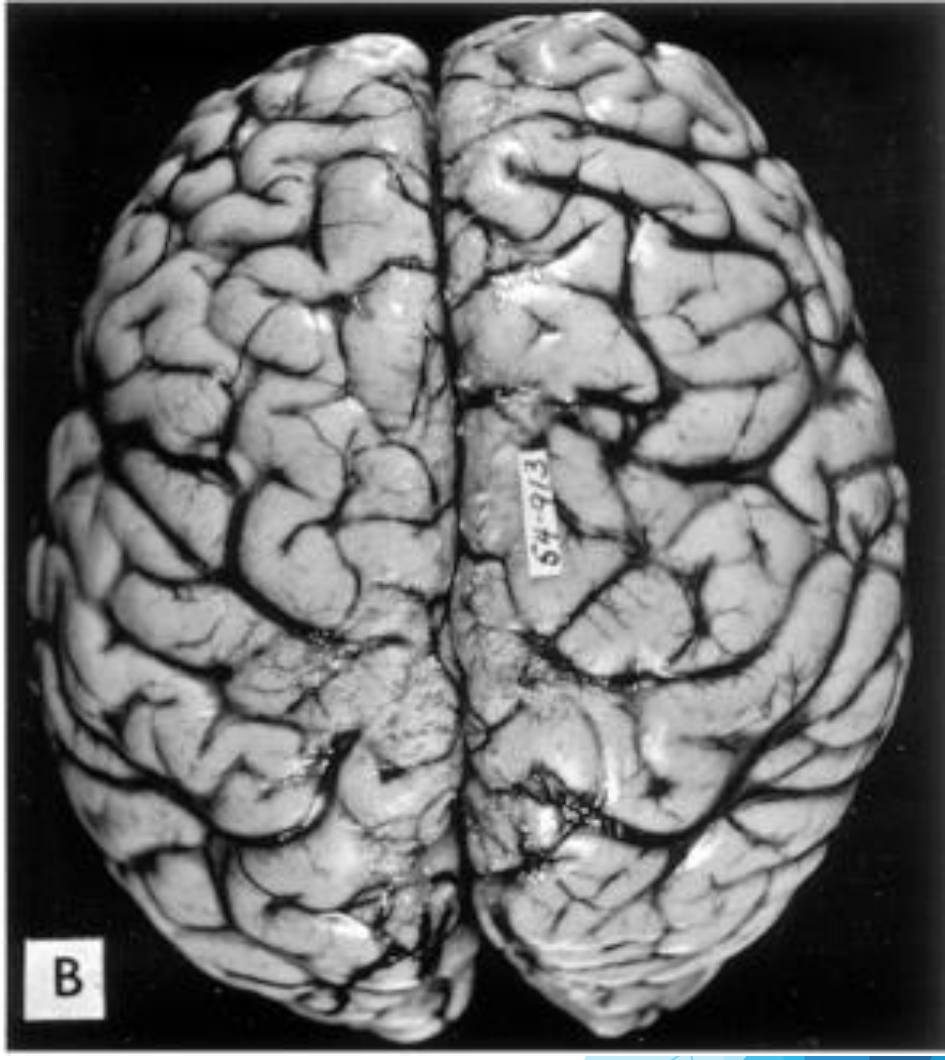
Left ventricular hypertrophy in patient with chronic hypertention

- **Atrophy:**
- is shrinkage in the size of the cell due to loss of cell substance.

Causes of atrophy are:

- 1.decreased workload.
- 2. loss of innervation.
- 3. diminished blood supply.
- 4. inadequate nutrition.
- 5. loss of endocrine stimulation.
- 6. aging.





Brain atrophy due to aging & reduced blood supply (atherosclerosis). Note that loss of brain substance narrows the gyri and widen the sulci.



Testicular atrophy

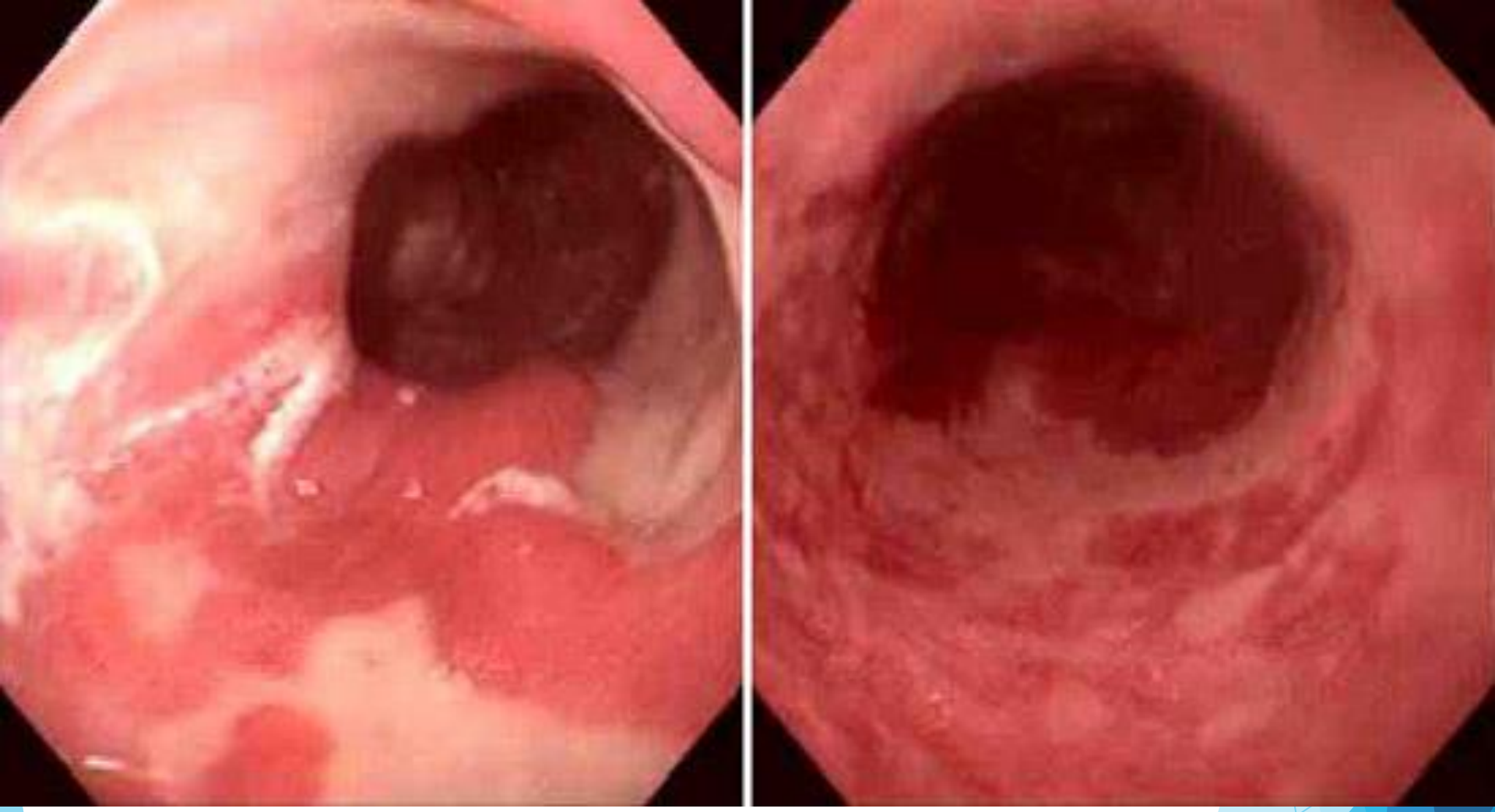
- **Metaplasia:**

is a reversible change in which one adult cell type is replaced by another (epithelial or mesenchymal).

The most common example is a change from columnar to squamous epithelium, as occurs in the squamous metaplasia of respiratory epithelium in response to chronic irritation.

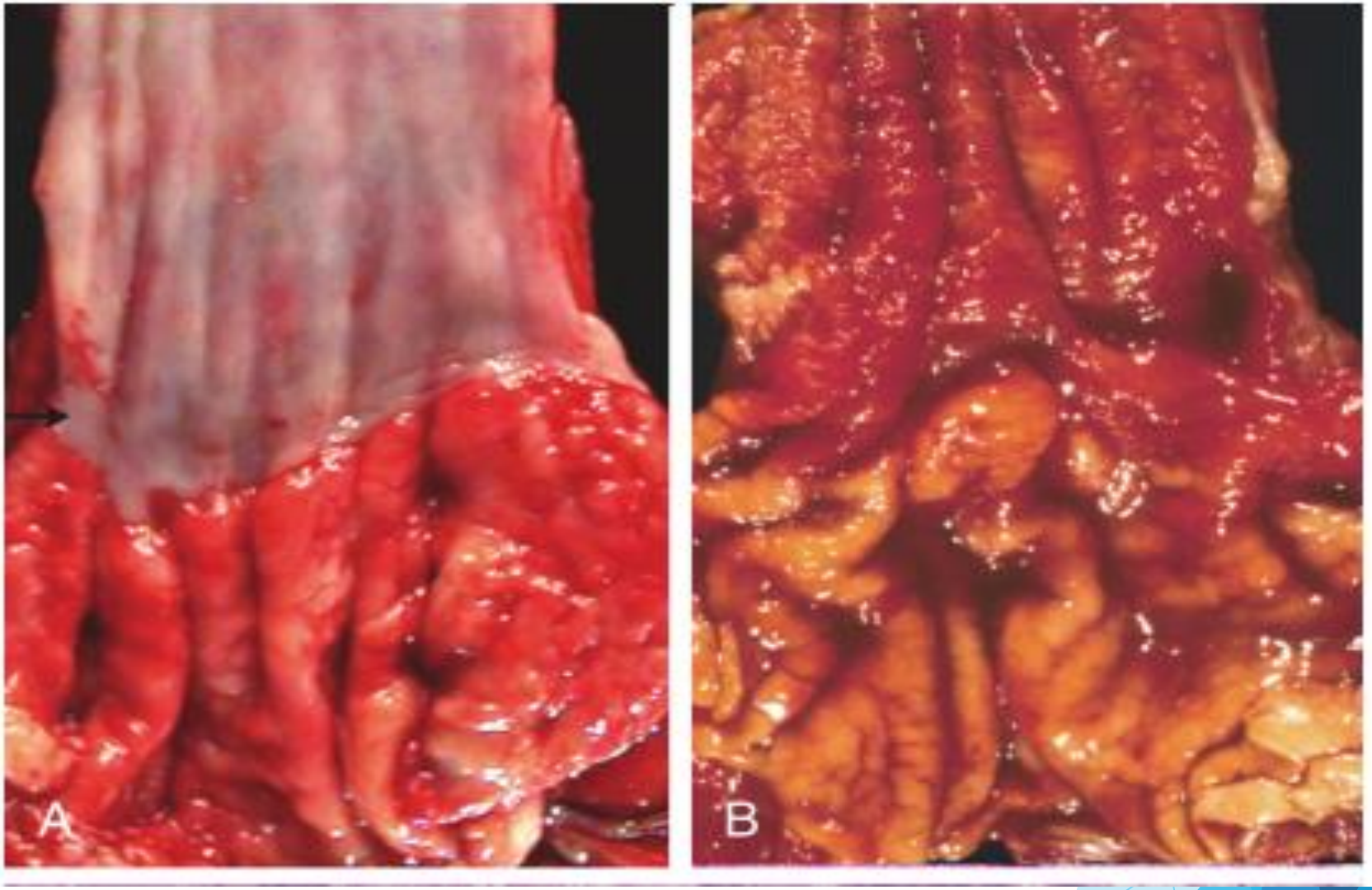
Although metaplastic epithelium is benign, the influences that predispose to such metaplasia if persist may induce atypical metaplasia which may progress to cancer transformation.

Metaplasia can occur in mesenchymal cells in which fibroblasts may become transformed to osteoblast or chondroblast to produce bone or cartilage.

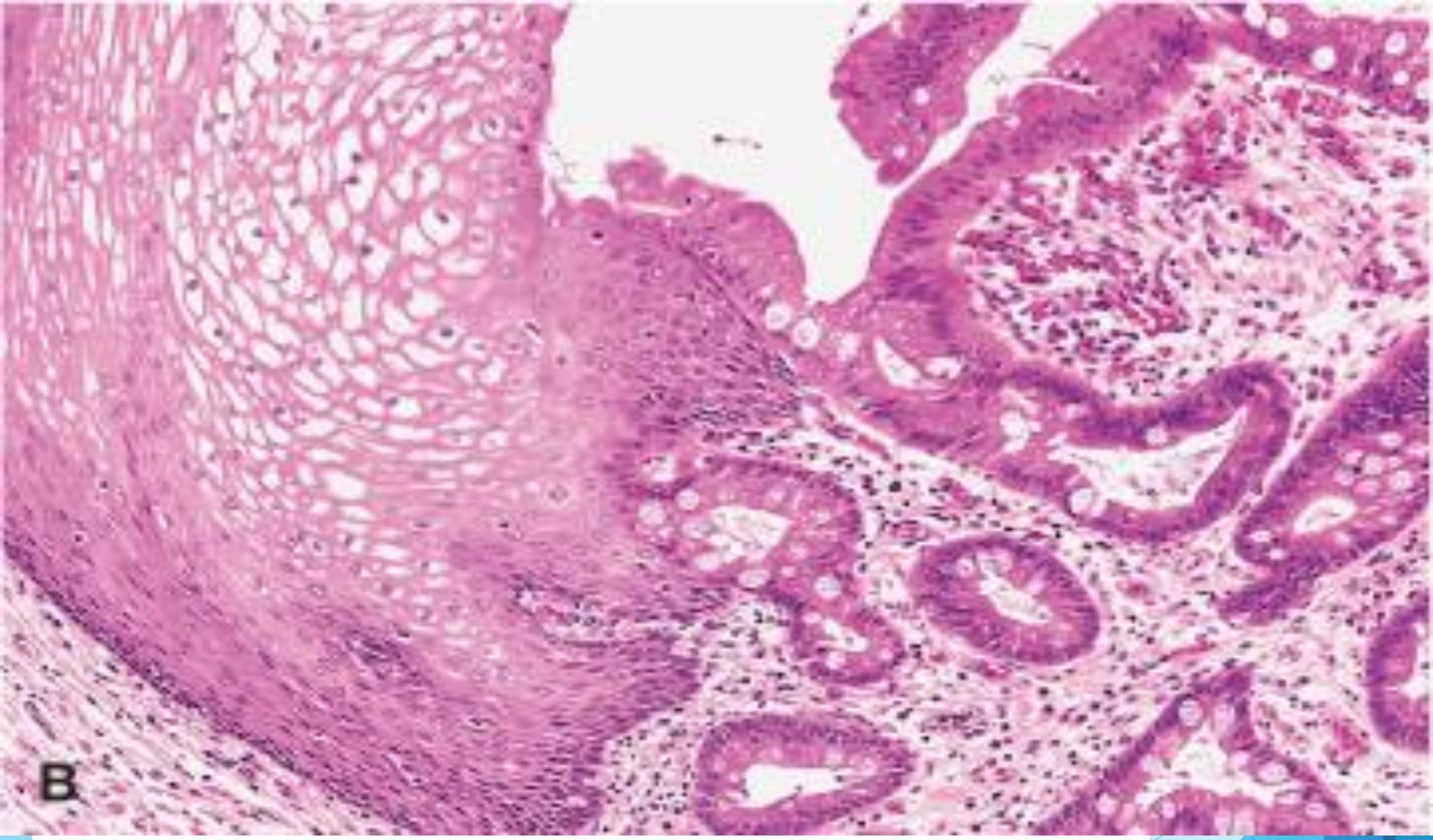


Endoscopical appearance of Barrett esophagitis

Squamous epithelium white in color while columnar epithelium red-pink in color.

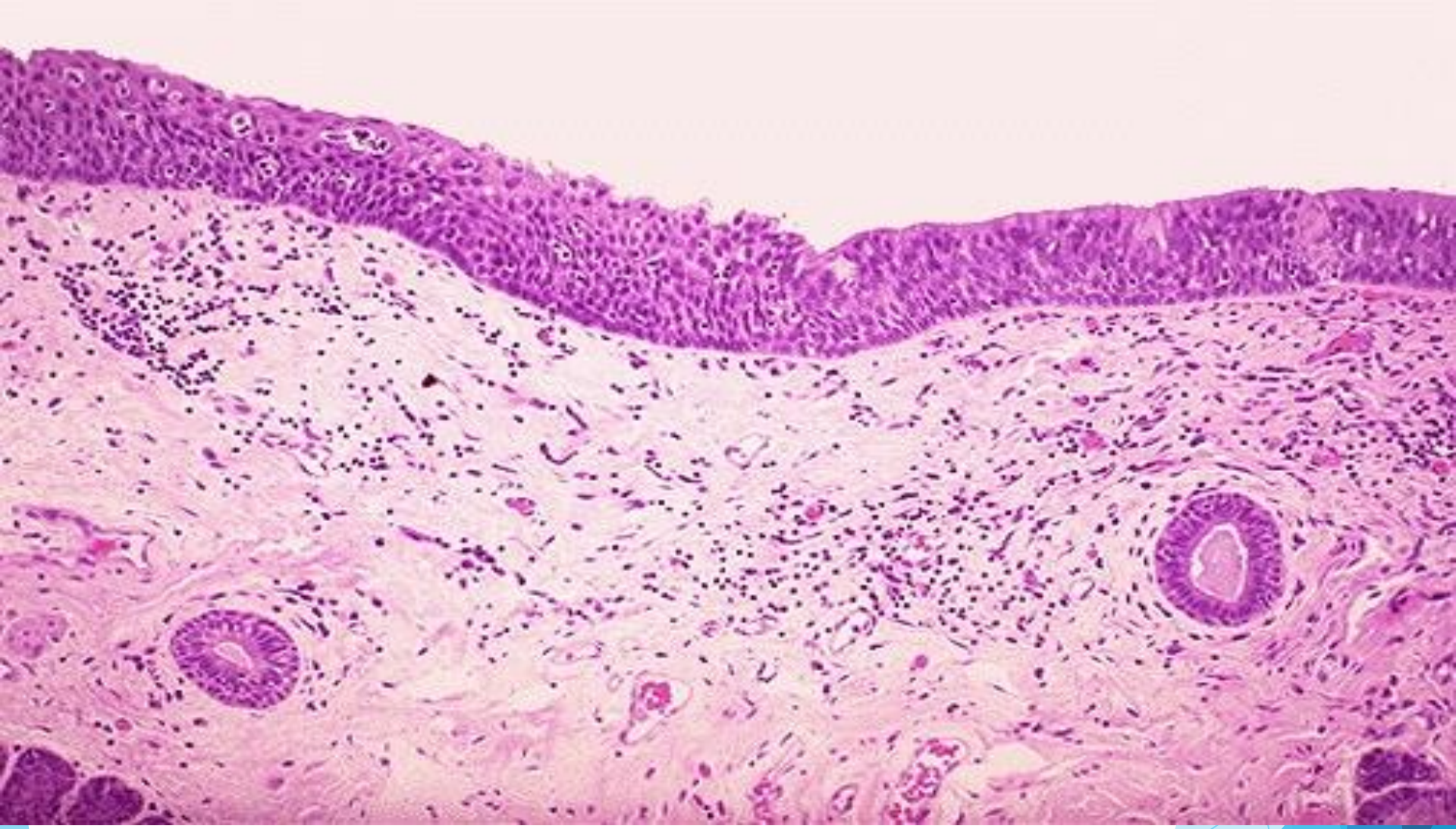


Barrett esophagus. A: Normal gastroesophageal junction. B: Note the small islands of paler squamous mucosa within the Barrett mucosa.



Barrett's esophagitis

(squamous to columnar metaplasia)



Squamous metaplasia of respiratory epithelium. Transformation of (pseudostratified ciliated columnar epithelium (Rt.side) to stratified squamous epithelium (Lt.side)

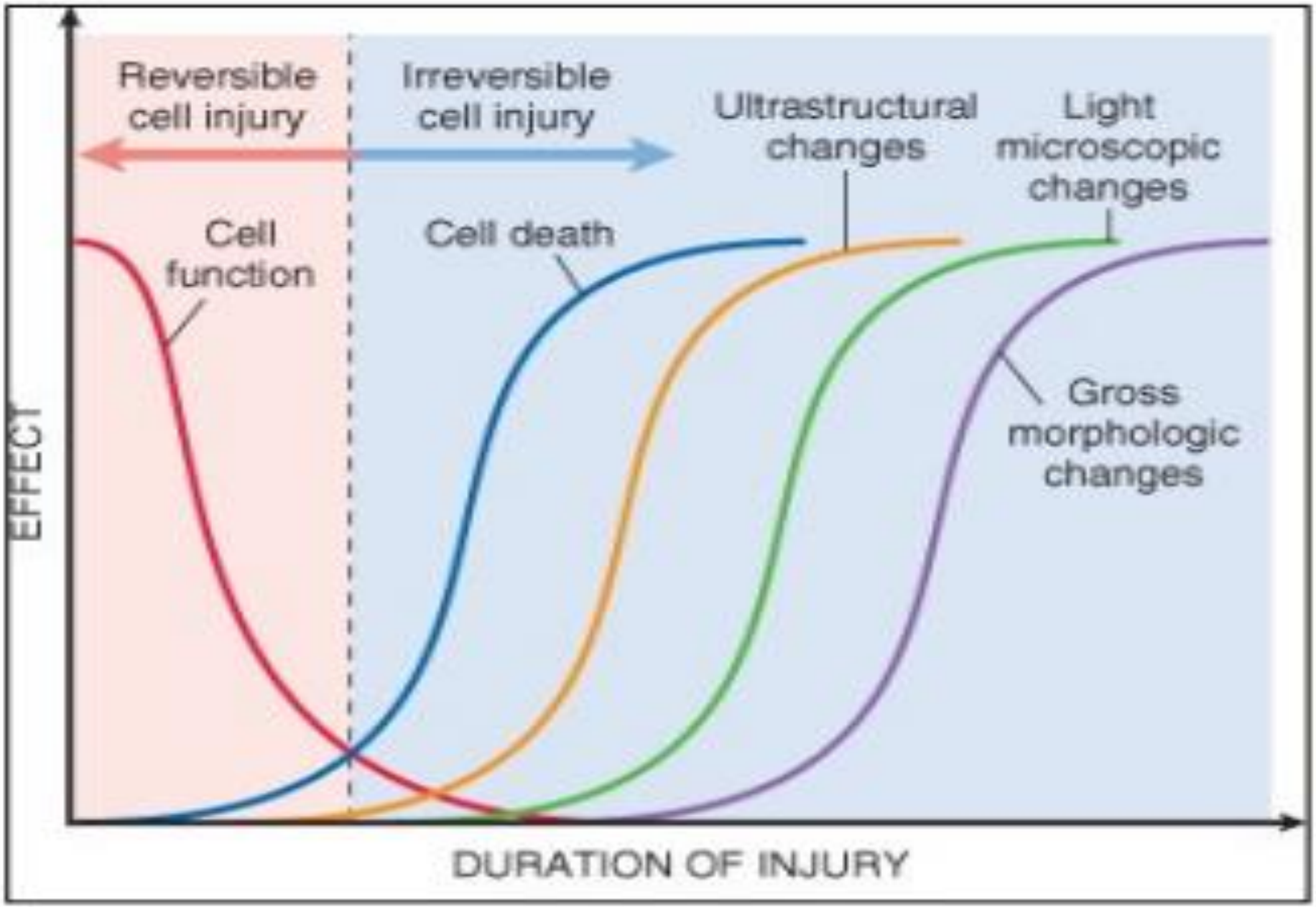
- **Reversible injury:**

means pathologic cell changes that can be restored to normalcy if the stimulus is removed or if the cause of injury is mild.

Mechanisms of cell injury

General Considerations:

- **Function is lost before morphological changes occur**
- **EM changes**
- **Microscopic changes**
- **Gross changes**


Reversible cell injury
Irreversible cell injury
Ultrastructural changes
Light microscopic changes
Cell function
Cell death
Gross morphologic changes
EFFECT
DURATION OF INJURY

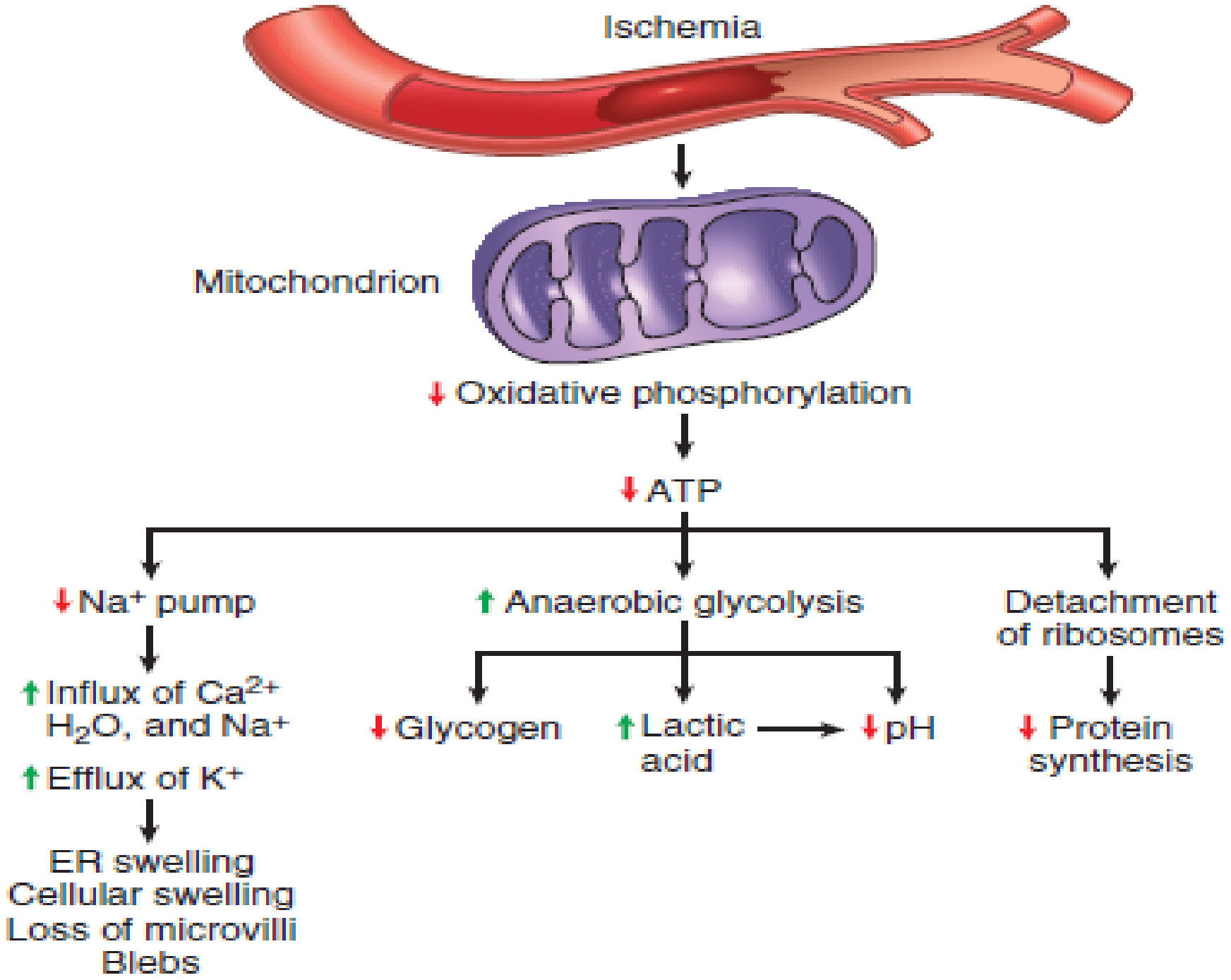
Mechanisms of cell injury

The biochemical mechanisms responsible for cell injury are complex but may be organized around a few relevant principles:

- Responses to injurious stimuli depend on the type of injury its duration, and its severity.
- The consequences of injury depend on the type, state, and adaptability of the injured cell.

Cell injury results from abnormalities in one or more of five essential cellular components:

- **1. Aerobic respiration, involving mitochondrial oxidative phosphorelation and ATP production.**
- **2. Maintenance of cell membrane integrity, critical for cell and organellar ionic and osmotic homeostasis.**
- **3. Protein synthesis.**
- **4. Intracellular cytoskeleton.**
- **5. Integrity of the genetic apparatus.**


Ischemia
Mitochondrion
↓ Oxidative phosphorylation
↓ ATP
↓ Na+ pump
↑ Influx of Ca2+ H2O, and Na+
↑ Efflux of K+
ER swelling
Cellular swelling
Loss of microvilli
Blebs
↑ Anaerobic glycolysis
↓ Glycogen
↑ Lactic acid
↓ pH
Detachment of ribosomes
↓ Protein synthesis

Morphology of Reversible Injury:

- Ultrastructurally :
- Generalized swelling of the cell and its organelles
- Blebbing of the plasma membrane
- Detachment of ribosomes from the endoplasmic reticulum
- Clumping of nuclear chromatin

Morphology of Reversible Injury:

- **Cell Swelling:**

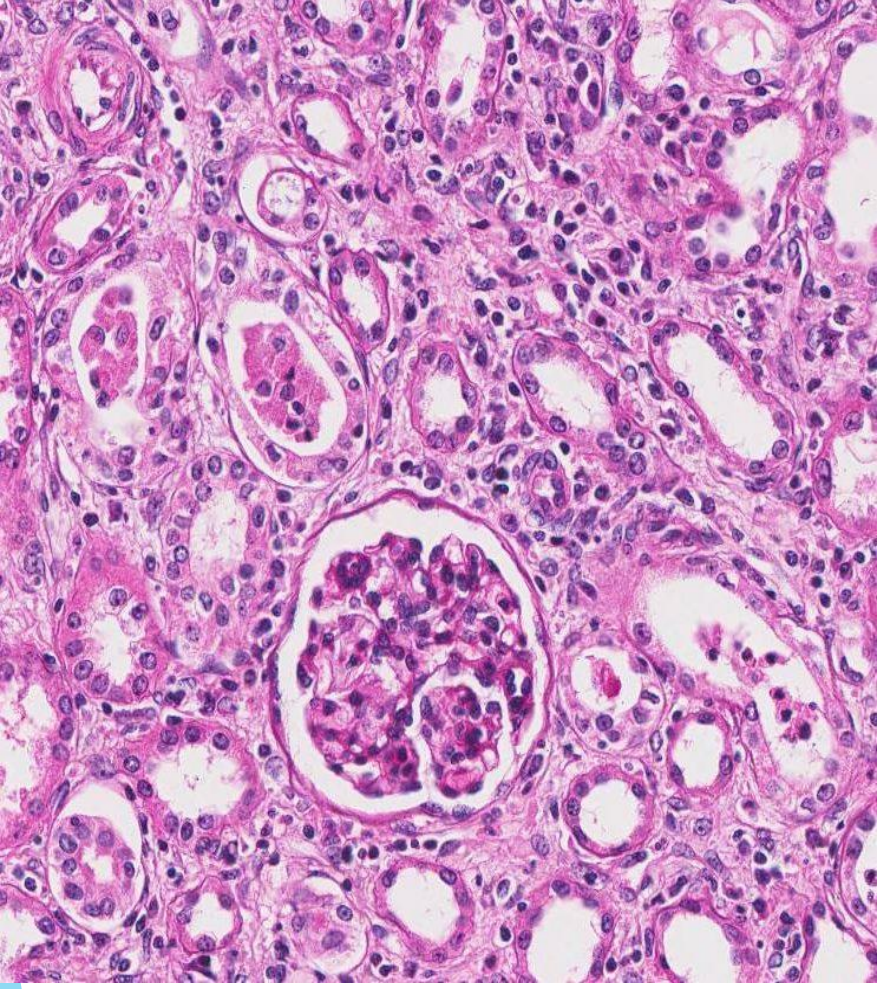
appears whenever cells cannot maintain ionic and fluid homeostasis (largely due to loss of activity in plasma membrane energy-dependent ion pumps).

- **It is difficult to appreciate with the light microscope;**

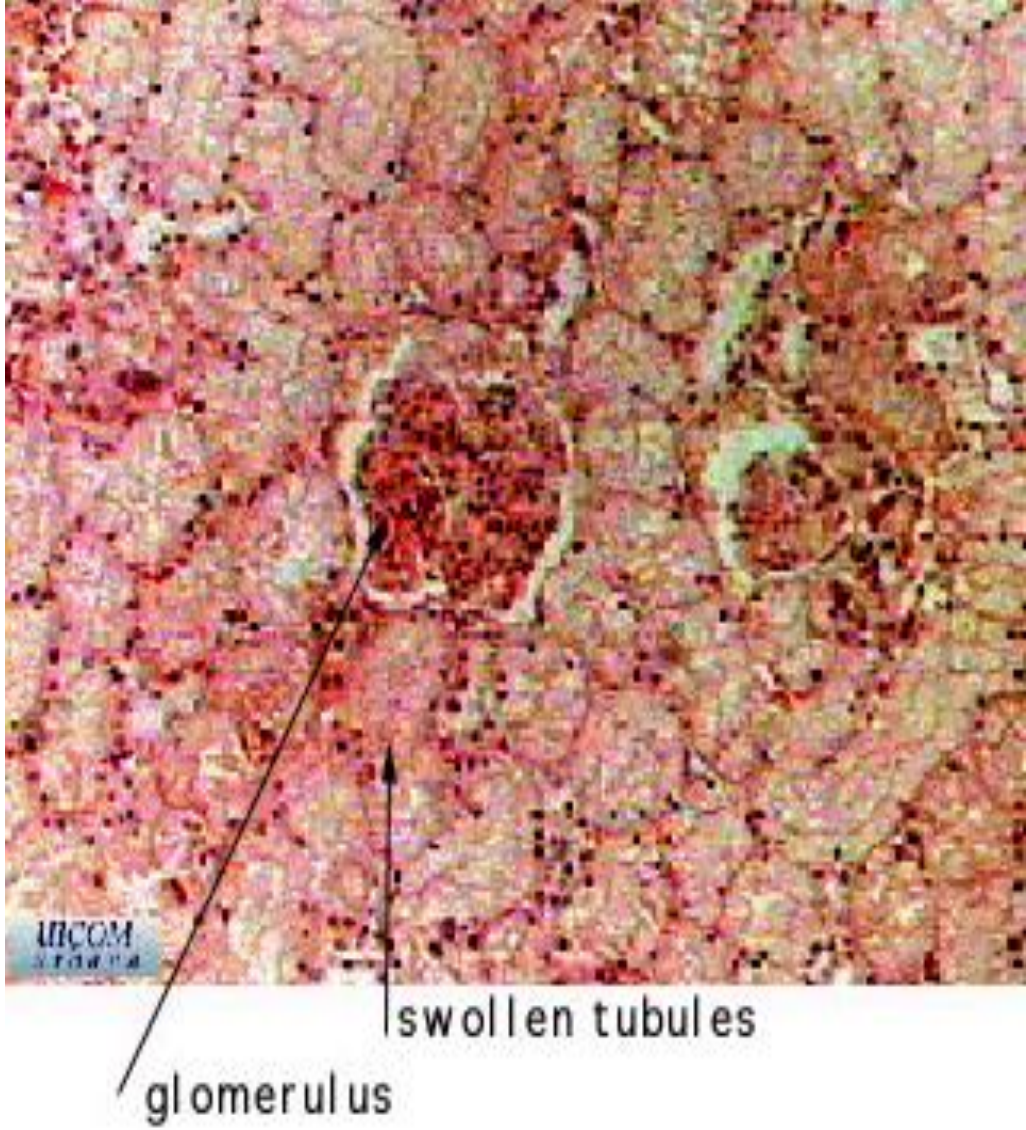
it may be more apparent at the level of the whole organ.
it causes some pallor, and increase in weight of the organ.

Microscopically, may reveal small, clear vacuoles within the cytoplasm.

This pattern of nonlethal injury is called *hydropic change* or *vacuolar degeneration*.
Swelling of cells is reversible.



Normal Kidney histology



Hydropic swelling of renal tubules

Fatty change:

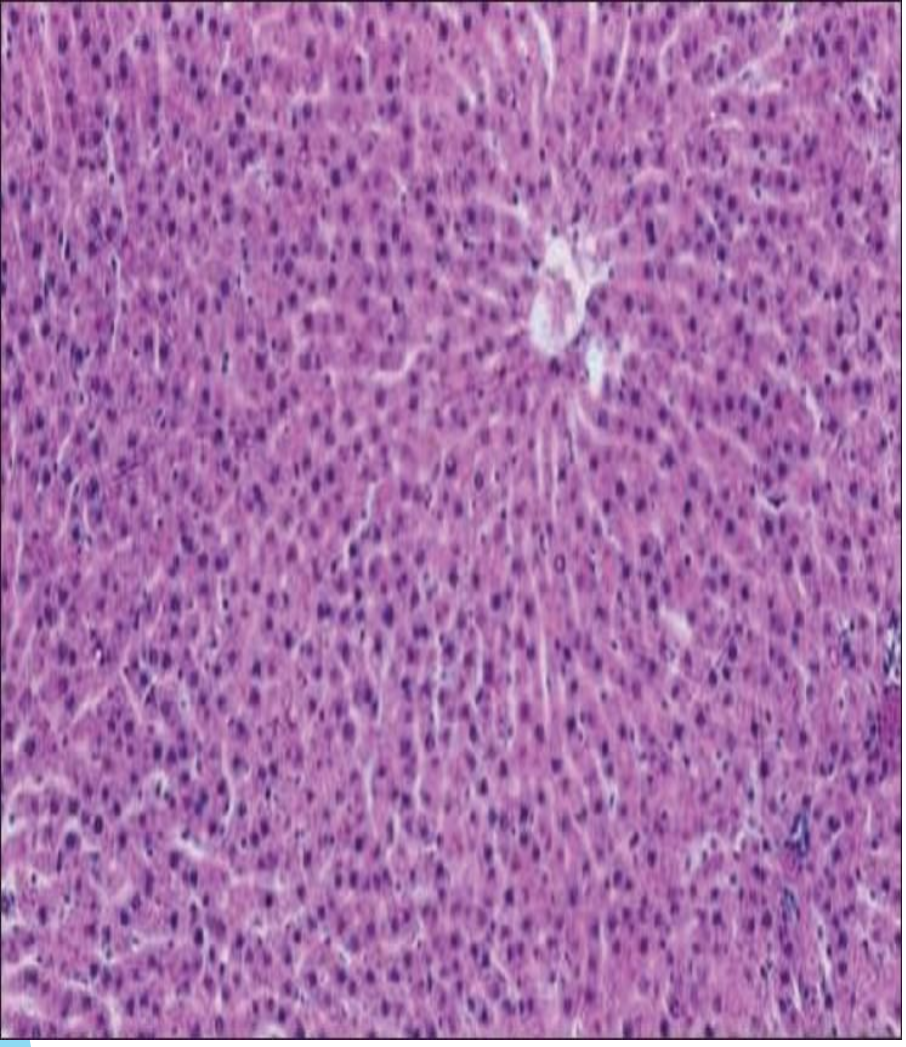
is manifested by cytoplasmic lipid vacuoles principally encountered in cells involved in or dependent on fat metabolism (e.g. hepatocytes and myocardial cells), and it is also reversible.

Normal liver

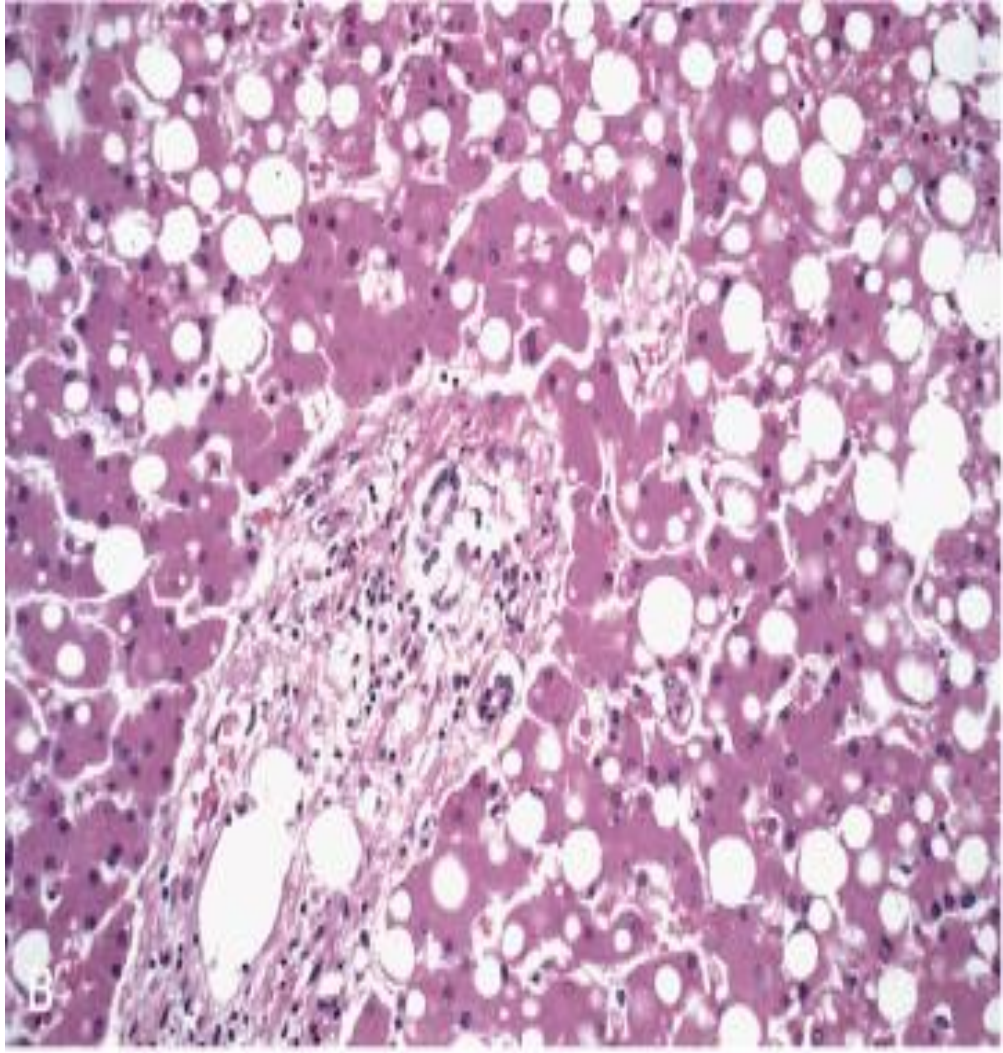




Fatty change of liver



Normal liver histology



Microscopical appearance of fatty change of liver

References:

1. Robbins and Cotran

PATHOLOGIC BASIS OF DISEASES

By : KUMAR – ABBAS – ASTER

2. Robbins

BASIC PATHOLOGY

By: KUMAR- ABBAS – ASTER

Thank You